

A hybrid global wolf pack algorithm-based incremental conductance method under partial shading conditions[☆]

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ABSTRACT

Solar photovoltaic (PV) power generation plays a crucial role in addressing energy shortages, promoting energy conservation, and reducing emissions. The efficiency and reliability of PV systems are enhanced by maximum power point tracking (MPPT) technology, which also helps to lower application costs. However, traditional MPPT methods often get trapped in local optima under partial shading, while intelligent methods fail to eliminate steady-state oscillations, causing unnecessary power losses. Therefore, a hybrid global wolf pack algorithm-based incremental conductance method (GWPA-FINC) is proposed to reduce oscillation along with improving the identification of the global maximum power point (GMPP) region under partial shading conditions. GWPA-FINC's performance is evaluated in seven scenarios: static and dynamic partial shading (scenario 1 to scenario 4), scenarios of varying temperature (scenario 5 to scenario 7). A comparative analysis is conducted with Particle swarm optimization (PSO), gray wolf optimization (GWO), sand cat swarm optimization (SCSO), wolf pack algorithm (WPA) and global wolf pack algorithm (GWPA), focusing on steady-state oscillation. Across all experimental scenarios, the power oscillation reduction up to 92.37% and voltage oscillation reduction up to 95.86% are achieved, with a tracking efficiency of 99.99%. These results indicate that the proposed GWPA-FINC effectively addresses the GMPP optimization challenge, significantly enhancing both steady-state performance and system efficiency under shading conditions.

1. Introduction

Energy challenges pose significant issues for humanity, and the development of renewable energy sources like solar power, has emerged as a critical solution to the energy crisis [1]. The efficiency of PV systems is closely linked to environmental conditions, particularly since uneven solar irradiance directly affects the output of PV cells, resulting in fluctuations in system performance. Consequently, improving the stability and efficiency of PV systems under partial shading through MPPT technology has become a central focus of research in the field of solar power generation.

Under uniform illumination, the power–voltage (P–V) characteristic curve of a PV array exhibits a single peak, with only one extremum—the GMPP. In such cases, traditional MPPT techniques are sufficient to track the maximum power point (MPP), offering both effectiveness and ease of implementation. Traditional MPPT methods include techniques based on the original system parameters, such as the fractional open-circuit voltage (FOCV) [2] and fractional short-circuit current (FSCC) [3,4] methods. They also encompass stepwise optimization

methods, including hill climbing (HC) [5–7], perturbation and observation (P&O) [8–10] and the incremental conductance method (INC) [11–13]. Because environmental change affects seriously system efficiency, it is essential to identify the changes. The sign of the product of the PV current and voltage increment ($dIdV$) was used to identify changes in irradiance [14,15]. This approach helped avoid misjudgment of INC under sudden increases in irradiance, although misjudgment in steady-state conditions could cause exit from steady state. The power difference (dP) between adjacent sampling periods was utilized to assess irradiance changes [16], but the threshold was challenging to adapt to various conditions. Additionally, a method has been proposed to eliminate steady-state oscillations by using the permissible error of $|I + VdI/dV|$ [17]. However, when partial shading occurs due to obstructions such as buildings, clouds, or trees, each PV module receives varying levels of irradiance. This variation causes the P–V curve to become multi-peaked, featuring multiple local maximum power points (LMPP) alongside one GMPP [18]. Traditional MPPT methods often

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struggle to track the MPP under these conditions, as they can become easily trapped in local optima, resulting in significant power losses.

To address the drawback of traditional MPPT methods, Intelligent MPPT methods blossomed, such as fuzzy logic control (FLC) [19–22], artificial neural networks (ANN) [23], and swarm intelligence (SI) optimization algorithms. SI algorithms, inspired by the predation behavior and role division observed in biological swarms, offer significant advantages in addressing complex problems and multi-objective optimization. Given that MPPT is fundamentally a nonlinear system parameter optimization problem, SI demonstrates considerable potential in optimizing multi-peak MPPT. Numerous researchers have integrated SI into MPPT strategies, including particle swarm optimization (PSO), gray wolf optimization (GWO), sand cat swarm optimization (SCSO), and the wolf pack algorithm (WPA). To mitigate the slow convergence and the tendency of PSO to become trapped in local optima, two linear factors guiding duty cycle updates were introduced into the optimization process alongside environmental change initialization [24]. The overall distribution MPPT algorithm was utilized to identify the GMPP region. After determining the initial value, PSO was employed for a precise local search to swiftly locate the GMPP [25]. Furthermore, an in-depth analysis of stability and steady-state behavior under partial shading conditions was conducted, leading to the proposal of a novel P–V curve convex area method. This method utilized two voltage boundaries to estimate the GMPP position, resulting in rapid convergence and minimal fluctuation during GMPP resolution [26]. A new MPPT technology based on SCSO is used to maximize energy generation in scenarios such as uniform irradiance, complex partial shading, step-varying stochastic irradiance, and gradual irradiance variations. Its rapid tracking capability, high efficiency, and effective energy collection contribute to the production of cleaner energy [27]. A comparative analysis of the performance of PSO, GWO, and the horse herd optimization algorithm under dynamic partial shading conditions revealed that the horse herd optimization algorithm had better tracking accuracy and faster convergence [28]. Other SI-based algorithms include ant colony optimization (ACO) [29,30], bat algorithm [31], Grass Hopper optimization (GHO) [32], and flower pollination algorithm [33]. Although SI algorithms offer certain advantages in multi-peak optimization and show promise for further development, MPPT requires not only superior tracking performance but also low complexity and feasibility. The complexity of intelligent MPPT algorithms is significantly higher than that of traditional methods, which poses challenges for practical implementation.

Hybrid MPPT Methods aim to combine the advantages of various methods. For instance, GWO was combined with P&O for efficient MPPT of a PV system subjected to rapid variation of solar irradiance and partial shading conditions [34]. The global search capability of modified krill herd algorithm was utilized to enhance FLC, thereby reducing the probability of FLC becoming trapped in local optima under shading conditions [35]. The combination of PSO and genetic algorithm (GA) was employed to address the MPPT challenges of floating PV systems [36]. A hybrid optimization controller combining GA and ACO was proposed, where the iteration parameters of the ACO were dynamically adjusted based on the slope of the P–V curve at the operating point, resulting in fewer iterations required for the MPPT method [37]. Hybrid methods capitalize on the strengths of various algorithms while mitigating their weaknesses, offering superior tracking performance and improved steady-state behavior compared to traditional MPPT methods.

The previously mentioned SI algorithms, such as PSO and GWO, exhibit specific strengths for MPPT across different shading scenarios. Nevertheless, these algorithms frequently struggle to reduce the steady-state oscillation. Thus, a holistic approach that reduces both power and voltage oscillation is essential. The WPA, has been applied to solve optimization problems across various fields like for addressing the multiple traveling salesman problem using a dual wolf pack search strategy [38], solving the camera network deployment problem for

Table 1

Parameters of 1Soltech 1STH-215-P PV Module at STC: $T = 25\text{ }^{\circ}\text{C}$, $S = 1000\text{ W/m}^2$.

Parameters	Label	Value
Maximum power	P_{MPP}	213.15 W
Maximum power point voltage	V_{MPP}	29.00 V
Maximum power point current	I_{MPP}	7.35 A
Open circuit voltage	V_{OC}	36.30 V
Short circuit current	I_{SC}	7.84 A
Open-circuit voltage/temperature coefficient	γ	$-3.6099 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$
Short-circuit current/temperature coefficient	α	$1.02 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$
The number of cells in series	N_s	60

visual coverage [39], and optimizing multi-peak MPPT in the field of solar power [40]. The use of WPA in MPPT control remains limited, suggesting opportunities for performance improvement. Consequently, an enhanced version of WPA (GWPA) designed for the initial phase of MPPT is introduced, enabling swift positioning in the vicinity of the global peak. To achieve steady-state oscillation-free GMPP under partial shading at the same time, an improved INC (FINC) is proposed. Therefore, a hybrid MPPT algorithm combining GWPA with FINC is proposed to reduce or even eliminate the power losses caused by steady-state oscillation while tracking the GMPP successfully. The main contributions of this paper are as follows:

1. Behaviors of fixed-step inferior avoidance pouncing, variable-step besieging, and leader hunting are proposed to enhance the local search capabilities and optimization accuracy of the WPA.
2. Strategies for eliminating steady-state oscillations and recognizing environmental changes are proposed to improve system efficiency and enhance the accuracy of environmental identification, respectively.

This paper is divided into four sections. In Section 2, a description of the PV system under partial shading conditions is provided. The proposed GWPA-FINC algorithm is introduced in Section 3. Simulation experiments to validate the performance of the adopted MPPT technique are detailed in Section 4. Finally, Section 5 presents the conclusions.

2. Establishment and characteristic analysis of PV module

The PV cell is a device that converts solar energy into electrical energy based on the fundamental principle of the photoelectric effect. The external characteristic model of a PV cell can be regarded as a parallel circuit consisting of a constant current source and a forward diode. The output characteristics are detailed as follows [41]:

$$\begin{cases} I = I_{ph} - I_d - I_{sh} \\ I_d = I_o \left[\exp\left(\frac{q(V + IR_s)}{AkT}\right) - 1 \right] \\ I_{sh} = \frac{V + IR_s}{R_{sh}} \end{cases} \quad (1)$$

where V is the output voltage of the PV cell, I is its output current, I_{ph} is the photoelectric current, I_o is the diode reverse saturation current, q is the electronic charge, A is the diode ideal factor, k is the Boltzmann constant, T is the absolute operating temperature of the PV cell, R_s and R_{sh} are the series resistance and parallel resistance, respectively.

The output power of an individual PV cell is limited. To meet specific voltage and power requirements during practical production, it is essential to combine multiple PV cell units into PV modules, arranged in both series and parallel configurations. Subsequently, these PV modules are combined to form a PV array, also configured in series and parallel. In this study, the PV module 1Soltech 1STH-215-P, modeled in Simulink, is employed as the research subject. The parameters under standard conditions are presented in Table 1.

Table 2
Irradiance of PV model under four scenarios.

Irradiance (W/m ²)	S1	S2	S3	S4	S5	S6
Scenario 1	1000	1000	1000	1000	1000	1000
Scenario 2	600	600	600	1000	1000	1000
Scenario 3	1000	1000	600	1000	1000	600
Scenario 4	1000	800	400	1000	800	400

Table 3
Temperature of PV model under three scenarios.

Temperature (°C)	T1	T2	T3	T4	T5	T6
Scenario 5	25	25	25	25	25	25
Scenario 6	65	65	65	25	25	25
Scenario 7	25	25	65	25	25	65

Table 4
Power extremes for four scenarios of varying irradiance.

Shading scenarios	GMPP voltage	LMPP1 voltage	LMPP2 voltage
Scenario 1	1265.74 W 86.34 V	–	–
Scenario 2	1017.60 W 86.56 V	–	–
Scenario 3	852.90 W 93.07 V	825.56 W 57.52 V	–
Scenario 4	703.40 W 58.47 V	578.51 W 93.89 V	363.88 W 28.91 V

Table 5
Power extremes for three scenarios of varying temperature.

Shading scenarios	Scenario 5	Scenario 6	Scenario 7
GMPP voltage	1265.74 W 86.34 V	1088.05 W 74.50 V	1193.60 W 81.03 V

It is generally known that irradiance significantly impacts the output characteristics of the PV module. The short-circuit current increases linearly with irradiance, while the open-circuit voltage exhibits only minimal variation. Higher irradiance levels lead to increased current and greater power output at the same voltage, with the maximum output power under adjacent irradiance conditions differing by several tens of watts. Under uniform irradiance, the PV module exhibits a single-peak characteristic.

Outdoor PV arrays can be partially obstructed by bird droppings, dust, branches, or low-hanging wires, leading to varying levels of irradiance across the array. To analyze the output characteristics of the PV array under various shading conditions, a 3×2 array model is established in Simulink. The irradiance levels of PV1 to PV6 are designated as S1 to S6, with temperatures set as T1 to T6. Four scenarios of irradiance variation are analyzed, with the irradiance settings detailed in Table 2. Scenario 1 simulates a non-shaded or uniformly shaded array, Scenario 2 represents vertical shading, Scenario 3 represents horizontal shading, and Scenario 4 represents varying levels of shading. The global maximum power, local maximum power, and corresponding voltages of the PV array under the four scenarios are detailed in Table 4. Three temperature variations are also analyzed, with the temperature settings specified in Table 3. Similarly, the global maximum power and corresponding voltage for the photovoltaic array under the three temperature conditions are provided in Table 5. Partial shading has a minimal impact on the system's open-circuit voltage, while the maximum output power of the photovoltaic array is significantly affected.

To achieve GMPPT control of multi-peaked characteristic curves under partial shading, an improved WPA is introduced in this study to identify the GMPP region. This approach is combined with an enhanced INC for local search, aiming to maximize output power while minimizing oscillations.

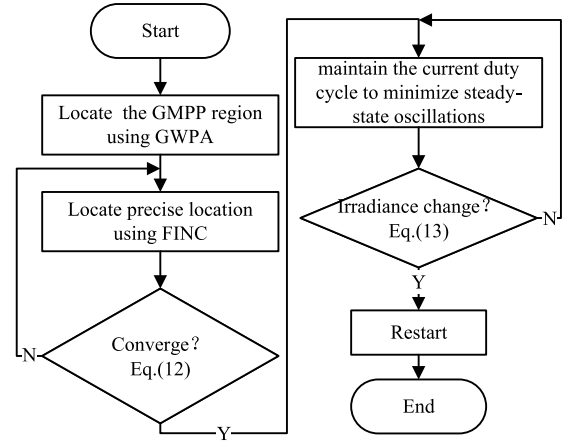


Fig. 1. Flowchart of GWP-INC.

3. The proposed GWP-INC algorithm

The GMPPT problem can be divided into two main tasks: locating the GMPP region and conducting a precise local search. Intelligent MPPT methods are capable of addressing the localization of the GMPP region, while traditional MPPT methods excel in fine-tuning the search within a unimodal curve, thereby minimizing oscillations. Therefore, in this study, a combination of an intelligent algorithm and the INC method is proposed. The global wolf pack algorithm (GWPA) is utilized to locate the GMPP region, and once the algorithm converges, the fine-tuned incremental conductance (FINC) method is initiated for precise local optimization. The steady-state convergence of the INC algorithm is determined by Eq. (11), maintaining the current duty cycle to minimize steady-state oscillations. Additionally, the detection of changes in environmental irradiance is performed according to Eq. (12), upon which the GWPA algorithm is restarted for further optimization, repeating the aforementioned process. The control logic of this process is illustrated in Fig. 1.

3.1. The mathematical model of WPA

WPA is an optimization algorithm inspired by the hunting behaviors of wolves. It employs a bottom-up design based on artificial wolves and a cooperative search structure, dividing the pack into leaders, scouts, and attackers [42]. The algorithm abstracts four intelligent behaviors: scouting, leader summoning, besieging, and survival of the fittest. The components of WPA are defined as follows:

(1) Initialization

Assume the number of variables to be optimized is D , representing a D -dimensional search space. The upper and lower bounds of the variables are denoted as X^U and X^L , respectively. The maximum number of iterations is K_m . Other parameters include the scout proportion factor α , the maximum number of scouting steps T_m , the distance determination factor ω , the step size factor S , and the update proportion factor β . Each artificial wolf's position represents a feasible solution, initialized as follows:

$$X_i^K = X^L + rand \times (X^U - X^L), i = 1, 2, \dots, n \quad (2)$$

where X_i^K denotes the position of the i th wolf in the K th generation, and $rand$ is a random number in the range $[0, 1]$.

(2) Leader selection criteria

The objective function is defined in Eq. (3), and the objective function value for each wolf is calculated based on its position. The wolf with the best value is designated as the leader, represented by the coordinates (X_{lead}, Y_{lead}) . If a wolf attains a better value, it is updated to

become the new leader. The leader guides the pack during summoning and besieging behaviors.

$$Y = f(X) \quad (3)$$

(3) Scouting behavior

After leader selection, the best-performing wolves are selected as scouts. The number of scouts S_{num} is randomly selected within the range $[n/(\alpha+1), n/\alpha]$ for each iteration. In the D -dimensional space, h forward directions are defined. Each scout moves one step in each of these directions with a fixed step size $step_a$. The scout i updates its position in the d dimension as follows:

$$x_{id}^p = x_{id} + \sin(2\pi \times \frac{p}{h}) \times step_a^d \quad (4)$$

where p is one of the h directions, and $d = 1, 2, 3, \dots, D$.

The best objective function value, Y_i , is compared with the leader's value, Y_{lead} . If $Y_i > Y_{lead}$, the scout is designated as the new leader, concluding the scouting process and initiating the leader's summoning. Conversely, if $Y_i < Y_{lead}$, the scout will continue scouting until either the maximum number of scouting steps, T_m , is reached or $Y_i > Y_{lead}$.

(4) Leader summoning

Wolves not classified as leaders or scouts are considered attackers. The leader summons all attackers to move towards its position with a larger step size, denoted as $step_b$. The position of the attacker x_j in the $m+1$ generation is updated as follows:

$$x_{jd}^{m+1} = x_{jd}^m + step_b^d \times \frac{g_d^m - x_{jd}^m}{|g_d^m - x_{jd}^m|} \quad (5)$$

where g_d^m is the prey's position in the d th dimension in the m th generation.

If any attacker achieves a superior objective function value $Y_j > Y_{lead}$, it is updated as the leader, and the leader summoning process is repeated. If $Y_j < Y_{lead}$, the attacker continues the pursuit until the distance d_{is} between it and the leader is less than the threshold d_{near} .

$$d_{near} = \frac{1}{D \times \omega} \times \sum_{d=1}^D |X_d^U - X_d^L| \quad (6)$$

(5) Besieging behavior

Attackers initiate a siege on the prey's position, represented by the leader's current position, using a smaller siege step size $step_c$. The position of attacker x_j during the siege is calculated as follows:

$$x_{jd} = x_{jd}^m + \lambda \times step_c^d \times |G_d^m - x_{jd}^m| \quad (7)$$

where λ is a random number uniformly distributed in the interval $[-1, 1]$, and G_d^m represents the prey's position in the m th generation. The relationship among the step sizes is typically as follows:

$$step_a^d = step_b^d / 2 = 2 \times step_c^d = |X_d^U - X_d^L| / S \quad (8)$$

where the step size factor S determines the step sizes in the WPA algorithm and influences the optimization accuracy.

(6) Updating mechanism

The wolf pack is updated based on the principle of "survival of the fittest", where the wolves with the lowest objective function values are eliminated and replaced with R new artificial wolves generated according to Eq. (2). The number R is randomly selected from the range $[n/(2 \times \beta), n/\beta]$, where β is the update proportion factor.

3.2. Global wolf pack algorithm (GWPA)

For an intelligent algorithm to successfully locate the GMPP, both excellent global search capabilities and robust local search abilities are crucial for achieving higher steady-state output power and minimizing output oscillations. However, traditional WPA face challenges such as low convergence accuracy and intense steady-state oscillation. To enhance the performance of WPA in GMPPT applications, this study proposes the following three improvements:

(1) Fixed-step inferior avoidance pouncing behavior

The exploration and pouncing behaviors of wolves in WPA involve multiple position updates per iteration. In the exploration phase, the maximum number of updates is determined by the maximum wandering steps T_m and the number of directions h . In the pouncing phase, the maximum number of updates is determined by the pouncing step size $step_b$ and the distance between the pouncing wolf and the leader. Consequently, the time complexity of WPA is difficult to estimate. For a pack of n artificial wolves, a complete iteration cycle may require $5n$ or more calculations, making the computation time several times more than that of algorithms like PSO or GWO for the same number of iterations.

In GMPPT applications, long-distance pouncing by wolves can lead to significant drops in PV output power, causing losses and consuming multiple algorithm cycles, which is not conducive to locating the GMPP. Therefore, in this study, the number of pouncing steps for wolves is capped, allowing each wolf to pounce a maximum of N_m times per iteration cycle, thus reducing time complexity and improving convergence speed. To fully utilize shared information among wolves, pouncing wolves also consider the information from exploring wolves: when a pouncing wolf x_j is close to the least favorable exploring wolf x_{wt} (i.e. $|x_j - x_{wt}| < d_{wt}$), it adopts a larger pouncing step $step_b$ to quickly exit regions with a lower probability of containing the GMPP. Outside the vicinity of x_{wt} , the pouncing wolf x_j resumes normal step sizes. This early termination of the pouncing behavior, coupled with smaller step sizes in the besieging behavior, helps avoid overshooting local maxima.

The pseudo code for this process is illustrated in Algorithm 1.

Algorithm 1 Fixed-step inferior avoidance pouncing.

```

for  $j = 1 : n - 1 - S_{num}$  do
  while The distance between the wolf  $x_j$  and the leader exceeds  $d_{near}$  do
    if  $|x_j - x_{wt}| < d_{wt}$  then
      the step  $step_b$  of the attacker  $x_j$  increase;
    end if
    if  $N < N_m$  then
      The wolf begins pouncing and updating the position using Eq. (5);
      if the objective function value of the wolf  $x_j$  is better than the leader then
        The wolf  $x_j$  becomes the new leader and begins summoning;
      Update the position of the leader;
      break;
    end if
     $N = N + 1$ ;
  else
    break;
  end if
end while
end for

```

(2) Variable-step besieging

In WPA, the primary function of besieging is local search, with a fixed besieging step size $step_c$. Despite the presence of random factors λ and distance difference $|G - x_j|$, theoretically, the besieging radius is $step_c \times |G - x_j|$, but only one besieging occurs per iteration cycle, which may not adequately cover the vicinity of the leader. In the early stages of the algorithm, when the wolf pack is dispersed, even after pouncing, not all wolves may be in the leader's vicinity. At this stage, a slightly larger besieging step (still smaller than the pouncing step) is advantageous for fine-tuning at the current positions of the pouncing wolves. Later in the algorithm, when the pack has converged, a smaller besieging step is beneficial for the final local search.

Thus, a variable-step pouncing behavior is proposed, where the besieging step size linearly decreases as iterations progress, starting from half of the maximum iterations $K_m/2$, with the initial step size set

at $step_c$, and decreasing to one-third of the initial step size by iteration K_m .

$$\begin{cases} step_c = k * \frac{K_m}{2} + b \\ \frac{step_c}{3} = k * K_m + b \end{cases} \quad (9)$$

The siege step for the K th ($K \geq K_m/2$) iteration cycle can be derived from Eq. (9) as:

$$step_K = -\frac{4step_c}{3K_m} * K + \frac{5}{3}step_c \quad (10)$$

(3) Leader hunting

In each iteration cycle of WPA, the leader itself does not participate in any intelligent behaviors and does not actively update its position unless replaced by a more optimal wolf. Although the besieging behavior searches the vicinity of the leader, only one besieging is performed per iteration cycle. Given that the leader's position represents the current optimal prey position, if it is near the GMPP, starting from the leader can lead to a quicker identification of a better prey position. Based on this, after the pouncing wolves complete their besieging, a leader hunting behavior is proposed, where the leader moves one step $step_d$ in each direction and compares the objective function values of all positions to find a better prey position, thereby improving optimization accuracy.

The pseudo code for the leader hunting behavior in GMPPT is illustrated in Algorithm 2.

Algorithm 2 Leader hunting.

```

for  $i = 1 : 2$  do
   $x_{hi} = G + (-1)^i * step_d$ ;
end for
Compare the objective function values of 3 positions;
Update the optimal position to the leader;

```

Since GMPPT does not require strong global search capabilities, and population updates can cause significant system fluctuations, the population update behavior has been discarded in the improved WPA.

3.3. GWPA for MPPT of PV systems

SI optimization algorithms in GMPPT are implemented primarily through two methods: using voltage as the particle position and using duty cycle as the particle position. In this study, the output voltage of the PV array is taken as the position of the artificial wolves, and PI control is used to regulate the voltage to the reference value corresponding to the artificial wolf. The input to the PI controller is the error between the reference value of the artificial wolf and the array voltage, and the output is the system's duty cycle. The objective function is based on the output power of the PV array, with the output voltage of the artificial wolf as the input. The flowchart of the proposed GMPPT-WPA (GWPA) method is shown in Fig. 2, with the following implementation steps:

(1) Initialization

The number of variables to be optimized is 1, i.e., the PV voltage, which takes values in the range of $[X^L, X^U]$; the maximum number of iterations is K_m ; exploration ratio factor α ; maximum wandering steps T_m ; distance decision factor ω ; step factor S ; maximum pouncing steps N_m ; and the initial positions of n wolves.

(2) Scouting behavior

The objective function value of each wolf is calculated, and the wolf with the best objective function value is designated as the leader (X_{lead} , Y_{lead}). The top S_{num} wolves with the best objective function values are selected as exploring wolves, while the remaining $n - 1 - S_{num}$ wolves are designated as pouncing wolves. In the GMPPT search space, two directions are defined (voltage increase or decrease). The exploring wolves wander with a step size $step_d$ for a maximum of T_m steps. If an exploring wolf's objective function value Y_i exceeds that of the leader

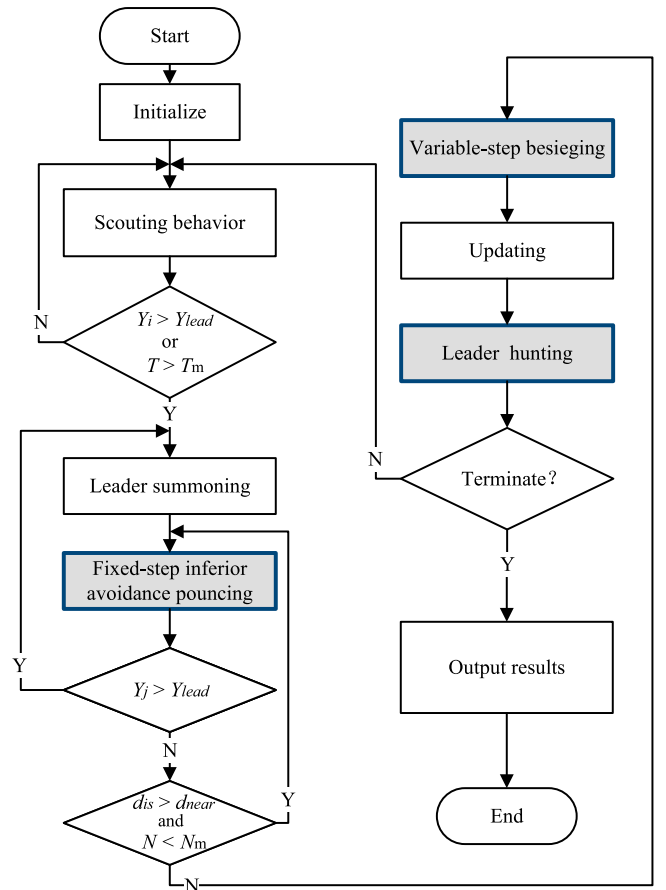


Fig. 2. Flowchart of GWPA.

Y_{lead} , the leader is updated, or if all wandering steps are completed, the process moves to step 3.

(3) Fixed-step inferior avoidance pouncing

The leader issues a call, and all pouncing wolves follow the fixed-step inferior avoidance pouncing principle, pouncing with a step size $step_b$. If a pouncing wolf's objective function value $Y_j > Y_{lead}$, the leader is updated, and the leader's position is re-announced. When the distance between the pouncing wolf and the leader $d_{is} < d_{near}$, or the maximum number of steps N_m is reached, the process moves to step 4.

(4) Variable-step besieging

The pouncing wolves besiege the prey position with a step size $step_K$. If $Y_j > Y_{lead}$, the leader is updated.

(5) Leader hunting

The leader hunts in two directions, comparing and updating the leader's information.

(6) Optimization accuracy check

If the preset optimization accuracy is met or the maximum number of iterations K_m is reached, the leader's position is output, and PI control adjusts the PV array voltage to the reference value; otherwise, the process returns to step 2.

3.4. Fine-tuned incremental conductance (FINC)

In the application of hybrid MPPT, the primary task of the traditional INC method is to perform a more refined search after locating the vicinity of the GMPP and to maintain the system parameters unchanged to reduce steady-state oscillations. Additionally, the method must accurately detect changes in irradiance conditions and promptly restart the intelligent algorithm as needed.

(1) Elimination of steady-state oscillations

The traditional INC method gradually optimizes, causing the output voltage and current to oscillate near the MPP during steady-state operation, leading to power loss and compromising the stability of the PV system. To minimize steady-state power loss and enhance system stability, it is crucial to reduce or even eliminate steady-state oscillations.

To eliminate steady-state oscillations in the PV system, an allowable error margin has been introduced into the traditional INC algorithm for evaluating the operating point. A lower bound for the step size of the INC is set, which acts as a fixed step size. This lower bound is determined by measuring the minimum slope value of $|I/V + dI/dV|$ near the MPP under uniform irradiance conditions, yielding the allowable error α . Since the minimum slope near the MPP varies under different irradiance conditions, α is compensated by adding a compensation factor δ to adapt to different environmental conditions.

When the system performs a local search using INC, and if the sampling results satisfy Eq. (11), the algorithm ceases perturbation, maintains the current duty cycle, and sets the steady-state flag to 1 ($Flag_s = 1$) until the irradiance changes.

$$|I/V + dI/dV| < \alpha + \delta \quad (11)$$

(2) Recognition of environmental changes

Fig. 3 illustrates the movement of the system's operating point when irradiance shifts abruptly from uniform shading Scenario 1 to partial shading Scenario 3. When tracking the MPP under fixed irradiance, the operating point moves along a fixed characteristic curve. In this case, the voltage and current increments in adjacent sampling periods must have opposite signs, i.e., $dIdV < 0$. However, when irradiance changes suddenly, the load curve remains unchanged at the instant of the change, causing the operating point to shift from A to B, where the voltage and current increments have the same sign, i.e., $dIdV > 0$. This characteristic is utilized to identify changes in irradiance [15]. However, during steady-state operation, the increments dI and dV caused by perturbation step size are close to zero, making the system susceptible to switching ripple, which can significantly influence the sign of dI and dV . Additionally, sampling errors may occur. These factors mean that even when the system is in steady-state, a condition of $dIdV > 0$ might appear, leading to inaccurate assessments of environmental changes and potentially causing the system to exit steady-state and drift away from the MPP.

Some studies have used the increment of PV output power to determine environmental changes during steady-state operation, based on the criterion that when the calculated result of sampled data at a given moment satisfies $|dP| > P^*$, a change in environment is identified, where P^* is a set power threshold (see, e.g., [43–45]). However, the challenge with this approach lies in setting P^* to accommodate different irradiance conditions.

Therefore, in this paper, a method combining $dIdV$ with dP is proposed to identify changes in irradiance. When the system is in steady-state and the sampling data satisfy Eq. (12), the algorithm will determine that the irradiance has changed, prompting a restart of the intelligent algorithm to re-locate the GMPP region under the new conditions. The flowchart of the improved free incremental conductance method (FINC) is shown in Fig. 4.

$$\begin{cases} dIdV > 0 \\ |dP| > P^* \end{cases} \quad (12)$$

4. Simulation results and analysis

To validate the effectiveness of the proposed method, a 3×2 PV array, as described in Section 2, is utilized as the study object. A simulation model based on a Boost circuit is constructed in MATLAB/Simulink, and the GMPPT control algorithm is implemented using the MATLAB Function module. Our experiments are conducted under ideal conditions that only considered partial shading. Throughout all

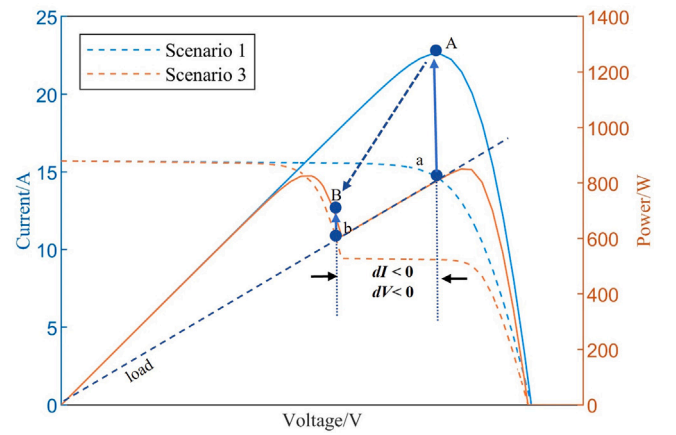


Fig. 3. Working point's movement after the change from Scenario 1 to Scenario 3.

experiments, the PV array is maintained at a constant temperature of 25 °C, employing the 1Soltech 1STH-215-P PV module, with parameters under standard test conditions listed in Table 1. The components of the Boost converter are configured as follows: $C_{in} = 1$ mF, $C_{out} = 5$ mF, $L = 0.5$ mH, $R_{load} = 50$ Ω , and the converter's switching frequency $f = 5$ kHz. The sampling period of the GMPPT module is set to 0.1 ms.

4.1. Performance of GWPA-FINC under static partial shading conditions

To assess the tracking performance of the proposed method GWPA-FINC under complex irradiance conditions with multiple peak characteristic curves, the four shading scenarios (Scenarios 1–4) designed in Section 2 are used as irradiance conditions. The tracking performance of PSO, GWO, SCSO, WPA, GWPA and GWPA-FINC algorithms is compared and analyzed. The power and corresponding voltages of GMPP and LMPP under the four scenarios are presented in Table 4. PI control is employed to regulate the PV voltage to the output reference value, with parameters $P = 0.0022$ and $I = 1.7$. For all algorithm implementations, the particle number $n = 20$, the maximum number of iterations $K_m = 15$, and the voltage range is set to $[0.1V_{oc}, 0.9V_{oc}]$, with particle positions initialized randomly. The PSO algorithm parameters are configured as $c = 0.5$, $a_1 = a_2 = 2$; for the WPA algorithm, the parameters are $\alpha = 4$, $T_m = 2$, $\omega = 50$, $S = 50$, $\beta = 4$, $step_a = step_b/2 = 2step_c = 0.8V_{oc}/S$, $d_{near} = 0.8V_{oc}/\omega$. The GWPA algorithm incorporate these WPA parameters with additional settings of $N_m = 3$, $step_d = step_c/4$. To meet the local optimization requirements, a fixed step size of $\Delta D = 0.001$ is selected for FINC.

1. Tracking performance under Scenario 1

In this scenario, characterized by the absence of shading, a single-peak characteristic curve is produced, achieving an ideal global maximum power of 1265.74 W at a voltage of 86.34 V. The tracking curves for each method are presented in Fig. 5, with steady-state performance in Table 6. The proposed GWPA-FINC method demonstrated the highest PV output power upon reaching steady-state, with an increase of approximately 4.12–5.55 W and a tracking efficiency of up to 99.95%. The power oscillation amplitudes observed for PSO, GWO, SCSO, WPA, and GWPA are 23.46 W, 24.52 W, 23.28 W, 29.72 W, and 19.12 W, respectively. Conversely, the GWPA-FINC method reduces the oscillation amplitude to 6.85 W, resulting in reductions of 70.80%, 72.06%, 70.58%, 76.95%, and 64.17% compared to the other five algorithms. Regarding voltage stability, GWO exhibits the highest voltage oscillation, which can negatively affect the system's performance. However, the GWPA-FINC method successfully minimizes voltage fluctuations, reducing the oscillation magnitude to 0.57 V. This reduction represents an impressive 89.84% improvement over the other algorithms. The other three methods, though effective in certain aspects, still produce

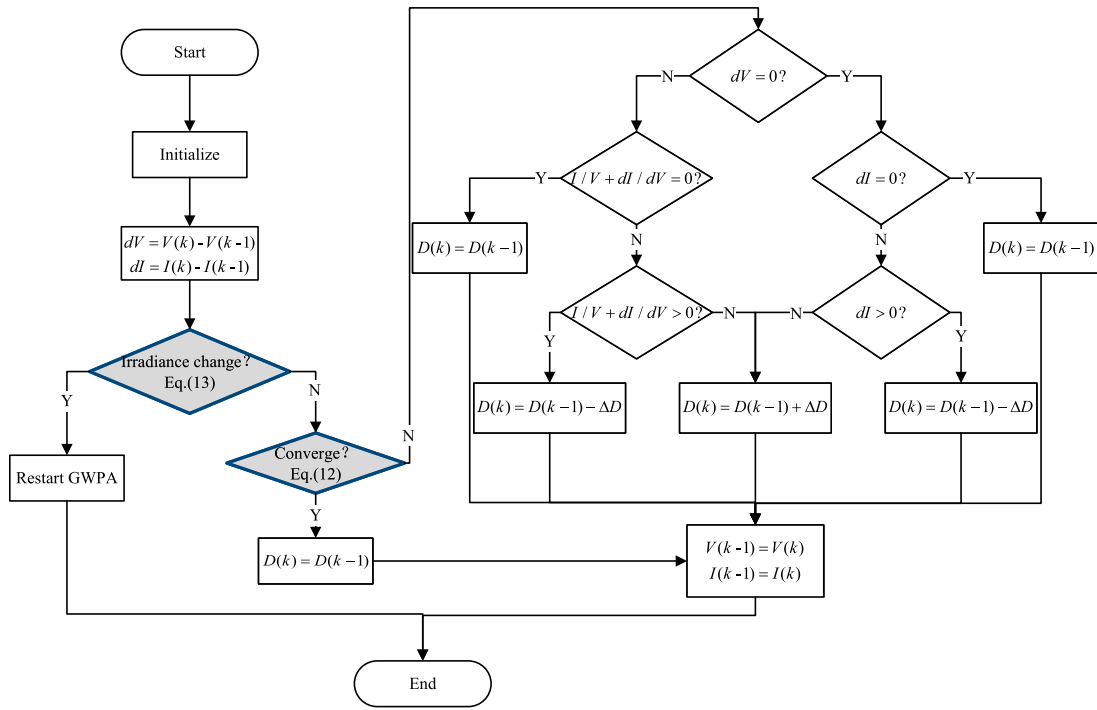


Fig. 4. Flowchart of FINC.

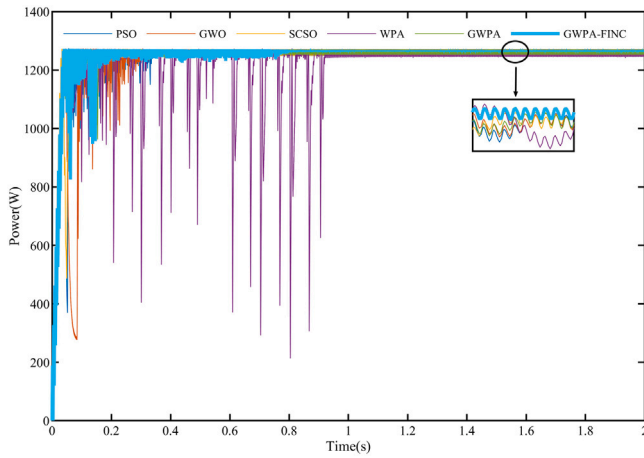


Fig. 5. Tracking curves for each method under Scenario 1.

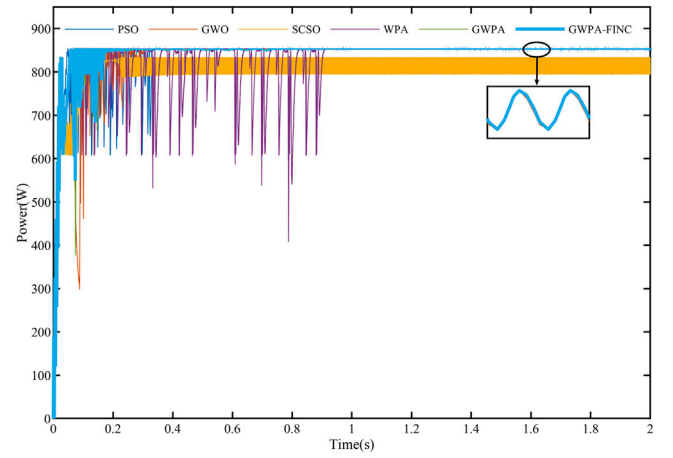


Fig. 7. Tracking curves for each method under Scenario 3.

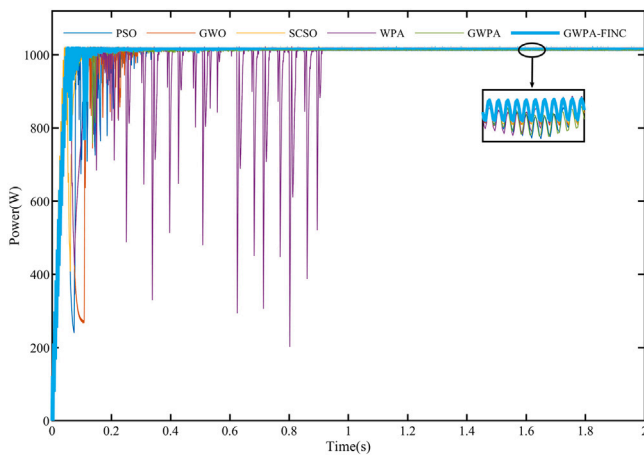


Fig. 6. Tracking curves for each method under Scenario 2.

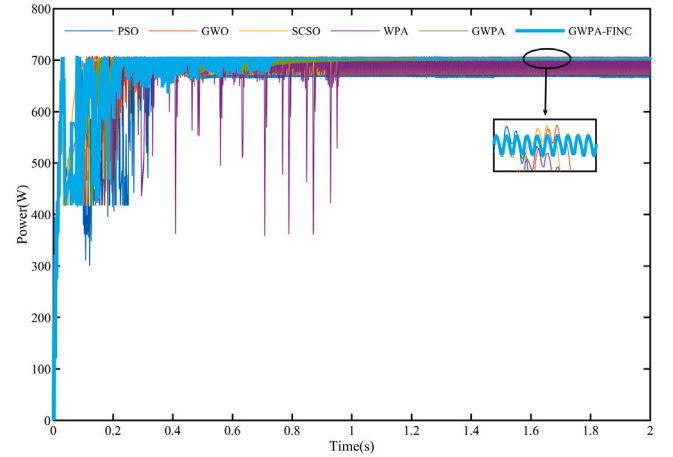


Fig. 8. Tracking curves for each method under Scenario 4.

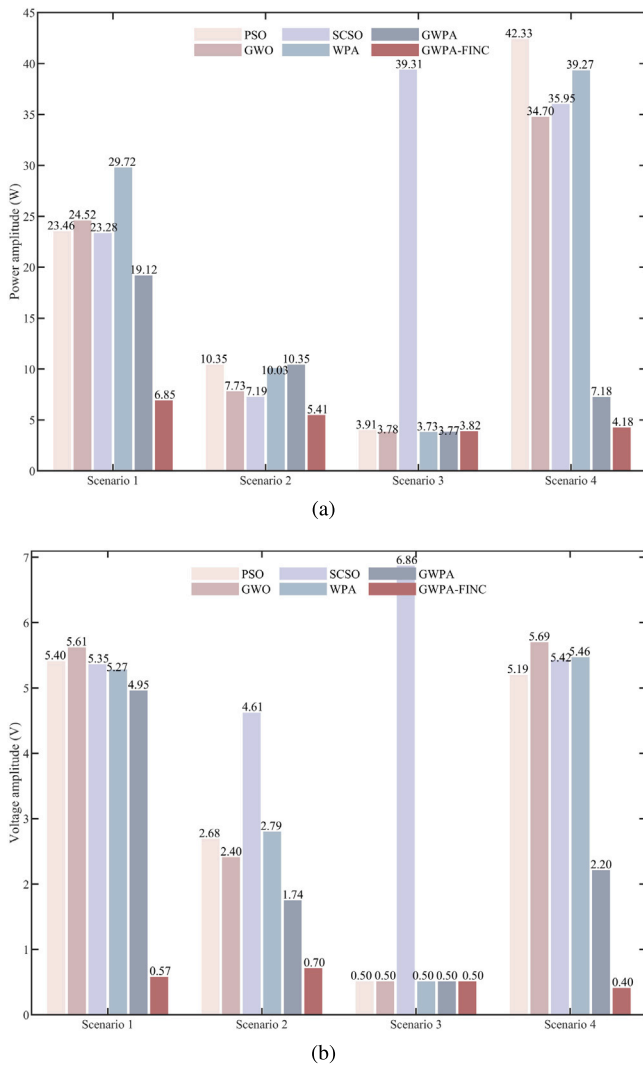


Fig. 9. Performance comparison of steady-state oscillations. (a) Power amplitude, (b) Voltage amplitude.

random voltage fluctuations, which could potentially lead to less stable system performance under Scenario 1.

2. Tracking performance under Scenario 2

This scenario involves longitudinal shading, producing a single peak characteristic curve with an ideal global maximum power of 1017.60 W at a corresponding voltage of 86.56 V. The tracking curves for each method under Scenario 2 are presented in Fig. 6, with steady-state performance listed in Table 7. The proposed GWPA-FINC method achieves a PV output power increase of approximately 0.94–1.42 W, with a tracking efficiency of up to 99.93%. Although SCSO has the highest tracking efficiency of 99.94%, its efficiency is only improved by 0.01% compared to GWPA-FINC. However, the voltage and power oscillations far exceed GWPA-FINC. On the one hand, the GWPA-FINC method decreases the power oscillation amplitude to 5.41 W, yielding reductions of 47.73%, 30.01%, 24.76%, 46.06%, and 47.73% relative to the other five algorithms. On the other hand, SCSO has the highest voltage oscillation, reaching 4.61 V, even GWPA with the smallest voltage oscillation has 1.74 V, while GWPA-FINC has the smallest voltage oscillation, only 0.70 V, which is at least 59.77% lower than the comparison algorithms.

3. Tracking performance under Scenario 3

Characterized by lateral shading, this scenario produces a dual-peak characteristic curve with an ideal global maximum power of

Table 6

Tracking performance under Scenario 1.

Methods	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
PSO	23.46	5.40	99.60%
GWO	24.52	5.61	99.62%
SCSO	23.28	5.35	99.64%
WPA	29.72	5.27	99.51%
GWPA	19.12	4.95	99.68%
GWPA-FINC	6.85	0.57	99.95%

Table 7

Tracking performance under Scenario 2.

Methods	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
PSO	10.35	2.68	99.83%
GWO	7.73	2.40	99.88%
SCSO	7.19	4.61	99.94%
WPA	10.03	2.79	99.86%
GWPA	10.35	1.74	99.88%
GWPA-FINC	5.41	0.70	99.93%

Table 8

Tracking performance under Scenario 3.

Methods	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
PSO	3.91	0.50	99.96%
GWO	3.78	0.50	99.98%
SCSO	39.31	6.86	95.65%
WPA	3.73	0.50	99.89%
GWPA	3.77	0.50	99.99%
GWPA-FINC	3.82	0.50	99.99%

852.90 W at a corresponding voltage of 93.07 V. The tracking curves for each method under Scenario 3 are shown in Fig. 7, with steady-state performance provided in Table 8. The proposed GWPA-FINC method increases PV output power by approximately 0.07–0.85 W, with a tracking efficiency of up to 99.99%. Except for the large voltage and power fluctuations in SCSO and the relatively low tracking efficiency compared to the other five algorithms, the performance of PSO, GWO, WPA, GWPA, and GWPA-FINC algorithms is almost the same.

4. Tracking performance under Scenario 4

This scenario features varying degrees of shading, yielding a triple-peak characteristic curve, with the GMPP positioned in the mid-voltage region, resulting in an ideal global maximum power of 703.40 W at a voltage of 58.47 V. The tracking curves for each method are shown in Fig. 8, accompanied by steady-state performance in Table 9. The GWPA-FINC method achieves a PV output power increase of approximately 0.64–10.78 W, with a tracking efficiency of up to 99.89%. At the same time, the power oscillation amplitude is reduced to 4.18 W, reflecting reductions of 90.13%, 87.95%, 88.37%, 89.36%, and 41.78% compared to the other five algorithms. As for steady-state voltage oscillation, GWPA-FINC still has the smallest oscillation amplitude, only 0.40 V, which is 81.82%–92.97% lower than the voltage oscillations of 5.19 V, 5.69 V, 5.42 V, 5.43 V, and 2.20 V of the other five algorithms.

Based on Fig. 5 to 8 and Table 6 to 9, it can be concluded that all SI algorithms possess the capability to handle partial shading conditions and locate the GMPP region. However, differences in local search capabilities result in varying convergence accuracies, leading to discrepancies in the obtained maximum power. Under the four shading scenarios, the proposed GWPA-FINC method consistently achieves the highest output power, enhancing system efficiency.

To enhance the comparison, bar graphs are created to show the results of power and voltage steady-state oscillation elimination for the five algorithms across four conditions, as illustrated in Fig. 9. The results indicate that under all four scenarios of varying irradiance, GWPA-FINC demonstrates the smallest power and voltage oscillation amplitudes at steady-state, resulting in the most stable output, aligning with the steady-state output stability sought in GMPPT. This composite

Table 9
Tracking performance under Scenario 4.

Methods	Power amplitude (V)	Voltage amplitude (V)	Tracking efficiency
PSO	42.33	5.19	98.36%
GWO	34.70	5.69	98.44%
SCSO	35.95	5.42	98.53%
WPA	39.27	5.46	98.41%
GWPA	7.18	2.20	99.80%
GWPA-FINC	4.18	0.40	99.89%

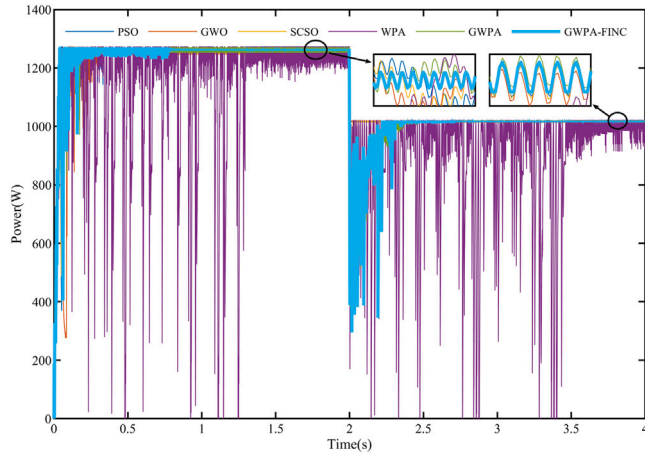


Fig. 10. Tracking curves for 6 methods during transition from Scenario 1 to Scenario 2.

method achieves higher output power and reduced output oscillations compared to GWPA alone under all irradiance conditions. This improvement is attributed to the fine search capability of the INC algorithm and the steady-state oscillation elimination strategy described in Eq. (11). The maximum steady-state output power increases by approximately 10.78 W, while power and voltage oscillations are significantly reduced, achieving a tracking efficiency of up to 99.99%. Therefore, the GWPA-FINC method, which exhibits minimal power loss and oscillation amplitude, is considered superior among the six methods.

4.2. Performance of GWPA-FINC under dynamic partial shading conditions

To further validate the effectiveness of GWPA-FINC, the ability of each method to manage partial shading transitions is analyzed. Irradiance conditions are set to transition from Scenario 1 to Scenario 2 and from Scenario 1 to Scenario 3. The tracking performance and steady-state behavior of the six algorithms for the transition from Scenario 1 to Scenario 2 are presented in Fig. 10 and Table 10. The corresponding performance during the transition from Scenario 1 to Scenario 3 is illustrated in Fig. 11 and Table 11.

From the perspective of handling partial shading transitions, the irradiance change detection condition derived from Eq. (12) effectively identifies the irradiance change and restarts the algorithm to reacquire the neighborhood of the GMPP.

In the single-peak curve tracking under Scenario 1, the maximum output power is increased by 0.98–22.87 W compared to other methods, while steady-state oscillations are reduced by 12.12–73.60 W, achieving a tracking efficiency of up to 99.96%. Under both Scenario 2 and Scenario 3, the proposed GWPA-FINC method consistently achieved the highest output power with the smallest steady-state oscillations. For example, in scenario 2, the oscillation amplitude is minimized to 4.96 W, which is 35.08%, 35.25%, 35.58%, 89.22%, and 24.73% lower than that of PSO, GWO, SCSO, WPA, and GWPA five methods, respectively. Even under Scenario 3, the steady-state power oscillation of GWPA-FINC at 3.73 W is slightly higher than

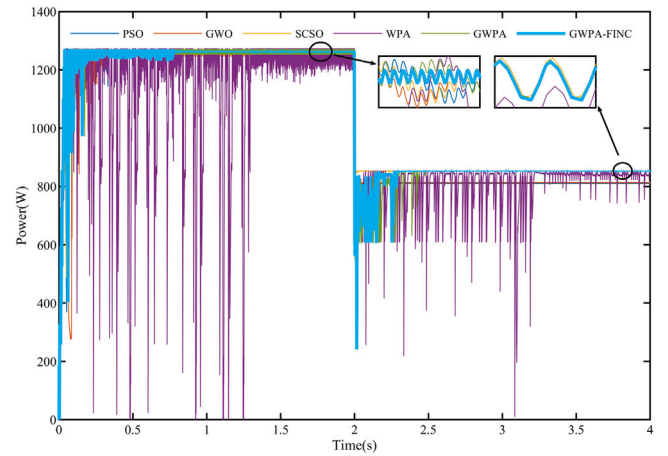


Fig. 11. Tracking curves for 6 methods during transition from Scenario 1 to Scenario 3.

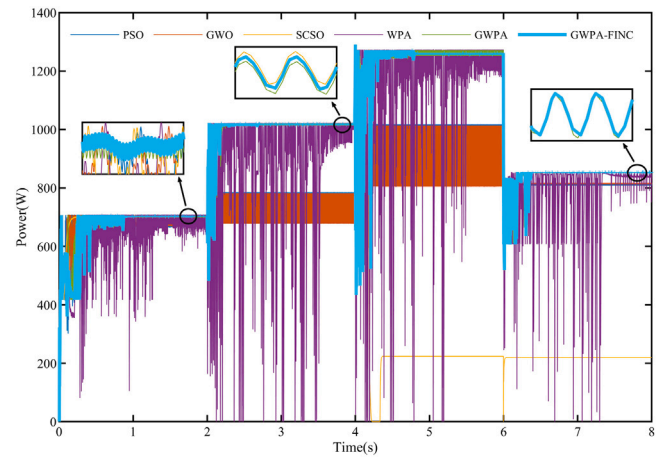


Fig. 12. Tracking curves for 6 methods when the solar irradiance varies rapidly.

that of GWPA at 3.71 W, and its steady-state voltage oscillation is slightly higher than that of SCSO at 0.49 V due to 0.50 V. However, its tracking efficiency is 99.99%, which is the highest among the six algorithms. Moreover, considering the overall situation of these two mutation scenarios, GWPA-FINC ensures tracking accuracy while reducing steady-state power and voltage oscillations. These results further verifies the effectiveness of the GWPA-FINC method in tracking the GMPP under scenarios of varying irradiance.

To assess the effectiveness of the proposed method under rapidly fluctuating solar irradiance, a dynamic condition is designed with sequential transitions from Scenario 4 to Scenario 2, Scenario 1, and finally Scenario 3. The power output curves for six methods are illustrated in Fig. 12, with tracking efficiency and steady-state oscillation statistics provided in Table 12. Fig. 13 employs a bar chart to visually compare the results. The findings reveal that GWPA-FINC consistently achieves the smallest steady-state oscillations and highest tracking accuracy during the first and second stages. In the third stage, PSO, GWO, and WPA display significantly higher steady-state oscillations compared to the other algorithms. Although SCSO eliminates power and voltage oscillations, it fails to track the MPP for the corresponding scenario, with tracking efficiency limited to around 20%. GWPA and GWPA-FINC yield comparable results in tracking efficiency and steady-state oscillations, but GWPA-FINC demonstrates superior overall performance by ensuring high tracking efficiency and minimizing steady-state oscillations.

Table 10

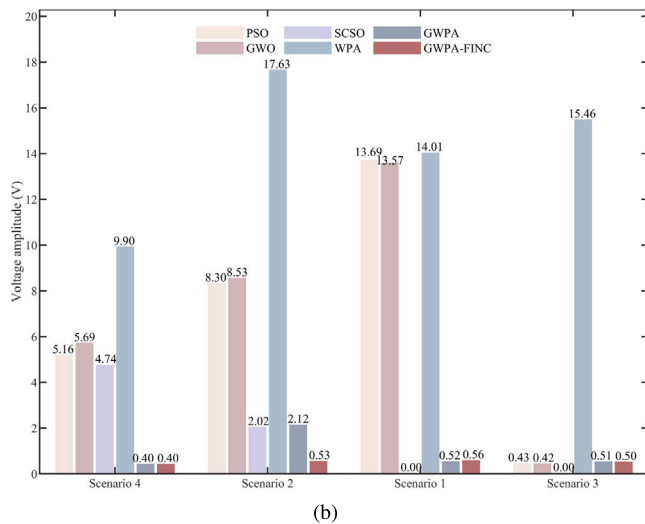
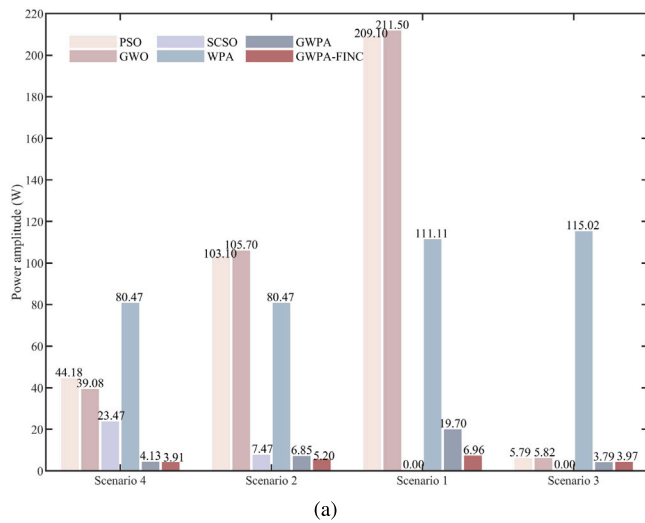
Performance of the five methods during transition from Scenario 1 to Scenario 2.

Methods	Scenario 1			Scenario 2		
	Power amplitude (V)	Voltage amplitude (V)	Tracking efficiency	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
PSO	23.03	5.32	99.63%	7.64	2.31	99.96%
GWO	25.28	5.21	99.57%	7.66	2.29	99.94%
SCSO	23.17	5.36	99.59%	7.7	2.43	99.95%
WPA	80.55	13.52	98.46%	46.03	9.66	98.98%
GWPA	19.22	4.89	99.72%	6.59	2.09	99.95%
GWPA-FINC	6.95	0.56	99.96%	4.96	0.53	99.97%

Table 11

Performance of the five methods during transition from Scenario 1 to Scenario 3.

Methods	Scenario 1			Scenario 3		
	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
PSO	22.86	5.29	99.64%	4.03	0.50	95.10%
GWO	25.14	4.98	99.65%	4.12	0.50	95.37%
SCSO	23.14	5.33	99.63%	3.73	0.49	99.98%
WPA	75.89	132.8	98.14%	33.04	7.25	98.99%
GWPA	19.18	4.90	99.87%	3.71	0.50	99.97%
GWPA-FINC	7.06	0.55	99.95%	3.73	0.50	99.99%

**Fig. 13.** Performance comparison of steady-state oscillations. (a)Power amplitude, (b)Voltage amplitude.**Table 12**

Tracking performance when the solar irradiance varies rapidly.

Scenarios	Methods	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
Scenario 4	PSO	44.18	5.16	98.41%
	GWO	39.08	5.69	98.49%
	SCSO	23.47	4.74	98.99%
	WPA	80.47	9.9	97.27%
	GWPA	4.13	0.4	99.67%
	GWPA-FINC	3.91	0.4	99.90%
Scenario 2	PSO	103.1	8.3	72.38%
	GWO	105.7	8.53	71.93%
	SCSO	7.47	2.02	99.96%
	WPA	80.47	17.63	98.37%
	GWPA	6.85	2.12	99.96%
	GWPA-FINC	5.2	0.53	99.98%
Scenario 1	PSO	209.1	13.69	72.70%
	GWO	211.5	13.57	72.43%
	SCSO	0	0.00	17.69%
	WPA	111.11	14.01	98.41%
	GWPA	19.7	0.52	99.69%
	GWPA-FINC	6.96	0.56	99.85%
Scenario 3	PSO	5.79	0.43	95.06%
	GWO	5.82	0.42	95.49%
	SCSO	0	0.00	25.80%
	WPA	115.02	15.46	96.97%
	GWPA	3.79	0.51	99.97%
	GWPA-FINC	3.97	0.50	99.98%

4.3. Performance of GWPA-FINC under varying temperature

In practical applications, in addition to partial shading causing irradiance variations and multiple local peaks, temperature fluctuations also affect the output power of PV arrays. To comprehensively evaluate the effectiveness and superiority of the proposed algorithm, three temperature variation scenarios (Scenario 5, Scenario 6, Scenario 7) are set, as shown in Table 4. The maximum power output and corresponding voltage for each scenario are displayed in Table 5. Two types of sudden changes are tested: from Scenario 5 to Scenario 6, and from Scenario 7 to Scenario 5. The tracking results for the six algorithms are presented in Figs. 14 and 15, while their tracking efficiency and steady-state oscillation data are summarized in Tables 13 and 14. From the comparison, it can be observed that, regardless of the scenario, the proposed GWPA-FINC algorithm achieves the highest tracking efficiency with the least power and voltage oscillation in the steady state. Under Scenario 6, although the voltage amplitude of PSO and GWO is nearly identical to that of GWPA-FINC, around 0.5 V, their power oscillations are significantly higher, and their tracking efficiency is only about 80%, failing to reach the maximum power output.

Table 13
Performance of the five methods during transition from Scenario 5 to Scenario 6.

Methods	Scenario 5			Scenario 6		
	Power amplitude (V)	Voltage amplitude (V)	Tracking efficiency	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
PSO	23.34	5.34	99.63%	16.28	0.52	80.55%
GWO	25.45	5.22	99.55%	15.92	0.51	79.64%
SCSO	23.28	5.32	99.59%	19.96	6.89	99.47%
WPA	73.08	13.52	98.28%	48.13	11.11	99.08%
GWPA	18.05	4.79	99.70%	18.05	6.12	99.63%
GWPA-FINC	6.63	0.56	99.78%	5.84	0.50	99.90%

Table 14
Performance of the five methods during transition from Scenario 7 to Scenario 5.

Methods	Scenario 7			Scenario 5		
	Power amplitude (V)	Voltage amplitude (V)	Tracking efficiency	Power amplitude (W)	Voltage amplitude (V)	Tracking efficiency
PSO	24.07	6.45	99.45%	84.02	7.97	97.10%
GWO	28.53	5.93	99.45%	71.32	7.35	97.65%
SCSO	24.33	6.44	99.51%	19.77	5.75	99.62%
WPA	81.13	11.83	98.53%	92.72	13.28	98.12%
GWPA	22.35	5.12	99.61%	18.05	4.66	99.70%
GWPA-FINC	7.07	0.56	99.95%	7.07	0.58	99.70%

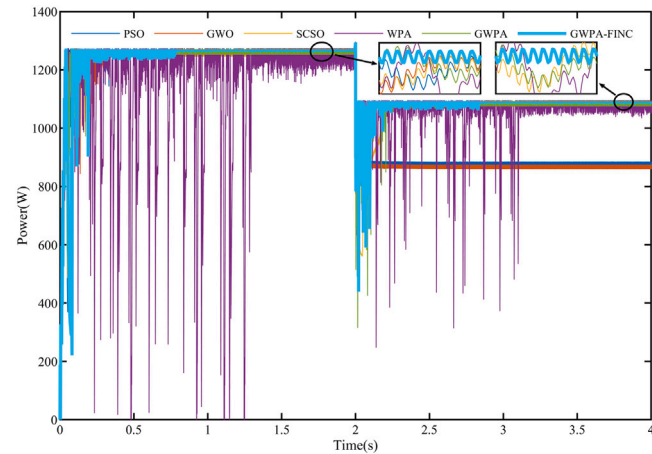


Fig. 14. Tracking curves for 6 methods during transition from Scenario 5 to Scenario 6.

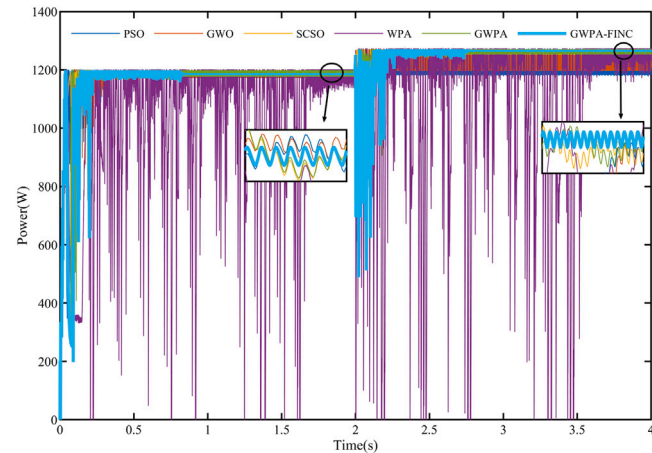
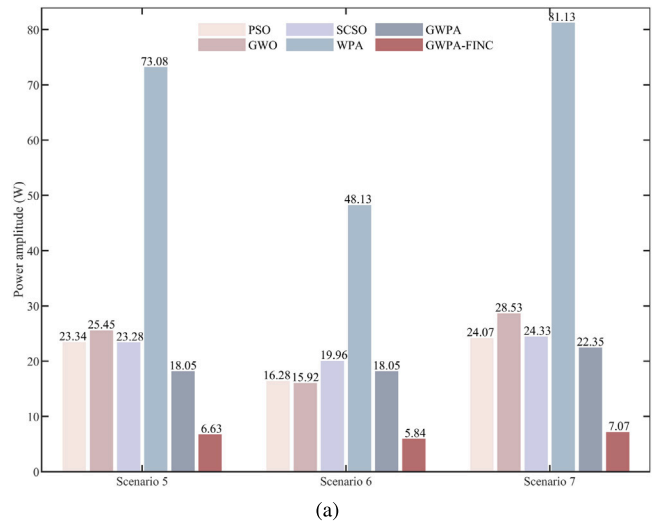


Fig. 15. Tracking curves for 6 methods during transition from Scenario 7 to Scenario 5.

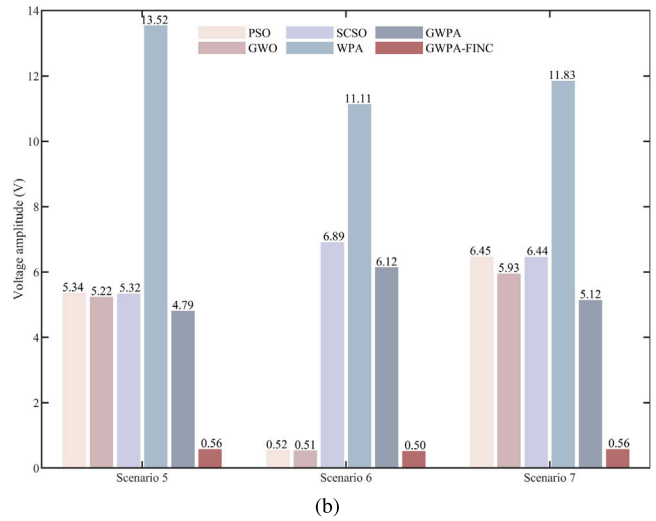


Fig. 16. Performance comparison of steady-state oscillations. (a) Power amplitude, (b) Voltage amplitude.

Table 15

Comparison of the proposed with existing MPPT algorithms.

Methods	Tracking efficiency	Steady-state oscillation	Parameter tuning	Training datasets	Adaptability	Complexity	Reference
P&O	Medium	Depend	No need	No need	Low	Simple	[8–10]
INC	Medium	Depend	No need	No need	Low	Simple	[11–13]
PSO	High	Medium	Need	Need	Medium	Medium	[26]
GWO	High	Medium	Need	Need	Medium	Medium	[28]
WPA	Medium	High	Need	Need	Medium	Medium	[40]
SCSO	Medium	High	Need	Need	Medium	Medium	[27]
ANN	High	/	Need	Need	Medium	Complex	[23]
FLC	High	/	Need	Depend	Medium	Complex	[19–22]
Proposed	High	Very low	Need	No need	High	Medium	This work

Overall, while the tracking efficiency improvement compared to the other five algorithms is not substantial, the steady-state oscillations are significantly reduced. The maximum reduction in power oscillation is 92.37%, with a minimum reduction of 60.83%. Voltage oscillation is reduced by 1.96% to 95.86%.

To provide a more intuitive comparison, the steady-state oscillations for the six algorithms are plotted as bar charts in Fig. 16. It is evident that the proposed GWPA-FINC method exhibits the smallest amplitude and the best performance. WPA, on the other hand, shows the largest steady-state oscillation amplitude, performing the worst. The other four algorithms exhibit similar steady-state oscillation amplitudes. In summary, GWPA-FINC delivers the best results, significantly reducing steady-state oscillation while ensuring accurate tracking.

Additionally, we propose a qualitative comparison between GWPA-FINC and metaheuristic MPPT techniques in terms of tracking efficiency, steady-state oscillations, parameter tuning, training datasets, adaptability and algorithm complexity, to highlight the uniqueness and advantages of the current work. Table 15 provides detailed comparative information of the proposed method with existing MPPT algorithms.

The results from all static shading, dynamic shading, and temperature variation experiments demonstrate that the proposed GWPA-FINC method significantly reduces steady-state power and voltage oscillations. This makes it a highly effective solution for optimizing PV system performance, especially in scenarios where steady-state stability and efficient tracking are crucial. The experimental results highlight the method's superiority in terms of both power output and system stability, providing a robust solution for real-world applications of MPPT under various conditions.

5. Conclusion

This paper presents a novel approach that integrates GWPA and FINC to achieve GMPPT with improved output power and reduced oscillations under variable shading conditions. To address the limited accuracy of the WPA algorithm, the GWPA method is proposed, which incorporates fixed-step pounce, variable-step siege, and leader wolf hunting behaviors to enhance optimization precision. By incorporating an enhanced INC method named FINC, steady-state oscillations are minimized, and environmental changes are effectively detected. Under static shading, as well as dynamic shading and temperature changing scenarios, the GWPA-FINC method significantly reduces steady-state oscillations and demonstrates excellent optimization accuracy. It outperforms all comparison algorithms in tracking accuracy and steady-state oscillation, reducing power oscillation by at least 24.73% and even 92.37% in some cases, with an average reduction of 75.24% in voltage steady-state oscillation. Additionally, the tracking accuracy of the GWPA-FINC method exceeds 99.89%, with some cases reaching 99.99%. Thus, the GWPA-FINC method effectively solves the multi-peak optimization problem under partial shading, leading to improvements in both system efficiency and stability. These results highlight the significant potential of the proposed method in enhancing the performance of PV systems. Given the impact and significance of our research in the industry and community, these improvements can

contribute to the widespread adoption of renewable energy and the optimization of energy systems, enhancing energy utilization efficiency and supporting sustainable development. Despite the promising results, limitations such as the absence of practical experiments due to hardware constraints must be acknowledged. Future work will focus on reducing transient oscillations, simplifying the GWPA-FINC method, evaluating its performance under a wider range of shading conditions, and preparing for practical applications. Extensive experimental testing will be conducted once suitable testing conditions become available. Moreover, our method is developed under ideal conditions, considering only partial shading and neglecting PV cell aging. Therefore, the potential impact of aging remains uncertain. Future work will incorporate real-world factors, such as aging, to assess their influence.

CRedit authorship contribution statement

Wenwen Xiao: Writing – review & editing, Software, Formal analysis. **Na Dong:** Supervision, Resources, Funding acquisition. **Kesen He:** Writing – original draft, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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